



		<u>Engineering Instruction 4/2</u>
<u>Title</u>		Date Issued : 20 August 2009
<u>Use Of Relief Chambers Fitted To Lindley Thompson Transformers</u>		Supersedes: None
Prepared By	S. Sabao (<i>O&CE</i>)	Review / Approved Date
Checked By	B. Siso (<i>DSM</i>)	
Approved By	J.M Chinembiri(<i>T&DD</i>)	

1. A number of 100kVA Lindley Thompson transformers are in service in the Commission and are fitted with a copper bellows type relief chamber in place of the conservator tank.
2. The following instructions should be noted and carefully observed.
3. A cover is placed over the abovementioned bellows and has a vertical slot through which the bellows can be seen. A temperature scale in degrees C. is marked up the side of this slot, showing the correct position of the top of the bellows for a range of oil temperatures.
4. To check that the oil Level in the transformer is correct, remove the bellows cover and slacken the plug in the top of the bellows to release any air which may have been trapped inside the transformer. As soon as oil begins to flow, tighten the plug and replace the cover. The temperature of the oil must then be measured. The thermometer pocket is situated in the centre of the blanking plate for the oil filler stand pipe. If the oil level is correct, then the top of the bellows should be level with the figure on the scale corresponding with the temperature of the oil.
5. If the temperature of the oil is 20°C, or less, the bellows will be almost fully contracted and there will be no indication as to the level of the oil. In this case, the transformer should either be left in the sun for several hours until the oil temperature is of such a value that the level can be checked as detailed in paragraph 5, or alternatively, if the oil temperature is 20°C 5%, the following procedure can be adopted.

Remove the bellows cover, the plug in the centre of the bellows, and the stand pipe blanking plate, Clamp the bellows between wooden blocks in the fully contracted position and fill up through the standpipe until oil appears through the plug hole, Replace the plug and tighten up; replace the standpipe blanking plate; remove the clamps and replace the cove. The top of the bellows should now be at the same level as the 20°C. mark on the temperature scale.

6. Since the oil in these transformers is completely sealed from contact with the air, there should be no deterioration of the electric strength or the acidity value.

Therefore, providing that no leaks develop and that the bellows remain at the correct height, oil samples should not be taken from these transformers during the three years period between one major overhaul and the next.

7. If the oil samples have to be taken while the transformer is in commission, then a shut down will have to be arranged and the same quantity of oil that has been taken out in samples must be put back into the transformer at once.
8. The following point might lead to some consternation if it were not appreciated. If the oil in these transformers is cold, and the bellows is fully contracted, there may be a considerable vacuum inside the tank, and, if, under these conditions, an attempt is made to run off a sample of oil, it will be found that no oil will flow out of the drain cock when it is opened. To obtain the sample, the vacuum will have to be broken by removing the standpipe blanking plate.
9. On no account should the bellows below be in the expanded position when the oil is cold, nor should they be contracted when the oil is hot. Obviously, in the latter case, a leak would be suspected.
10. In an emergency, if the transformer has to be topped up when the oil is hot, measure the oil temperature and the distance between the lid of the transformer tank and the figure corresponding to this temperature on the scale on the bellows cover. Remove the cover, wedge the bellows open slightly with wooden blocks, remove the centre plug and fill with oil through this plug hole, gradually wedging the bellows further and further open as the oil level comes up, until the top of the bellows is the correct distance from the lid of the transformers tank, Then reassemble.

Note: Excessive force must not be used to wedge the bellows open. The operation should be carried out in small steps.

